



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 12/01/2023*

PROBLEM - We observe N samples of the following discrete-time real-valued w.s.s. random process:

$$X[n] = S[n] + W[n] = A + W[n], \quad n = 0, 1, 2, \dots, N-1,$$

where A is a constant signal modelled as a Rayleigh distributed random variable:

$$f_A(a) = \frac{2a}{\sigma_A^2} e^{-\frac{a^2}{\sigma_A^2}} u(a), \quad \text{with } E\{A\} = \sqrt{\frac{\pi}{4}} \sigma_A, \quad E\{A^2\} = \sigma_A^2,$$

and $W[n]$ is additive white Gaussian noise (AWGN), zero mean, with power σ_W^2 .

Derive:

- (1) the signal to noise power ratio SNR $\gamma = E\{S^2[n]\} / E\{W^2[n]\}$, the autocorrelation function (ACF), and the Power Spectral Density (PSD) of the observed process $X[n]$;
- (2) the mean vector $\mathbf{m}_X$ and the covariance matrix $\mathbf{C}_X$ of the observed data vector:
 $\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T$;
- (3) the Maximum A Posteriori (MAP) estimator of the signal of interest A given the observation of $\mathbf{X}$ and express it as a function of the SNR , of the sample size N , the noise power σ_W^2 , and of the ML estimator of A ; discuss what happens when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (keeping constant σ_A^2);
- (4) the Bayesian Cramér-Rao bound (BCRB) and plot it as a function of the SNR and of the sample size N ; discuss what happens when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (keeping constant σ_A^2).

Hint_1: $ax^2 + bx + c = 0 \rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Hint_2: Use the following approximation: $E\left\{\frac{1}{a^2}\right\} \cong \frac{1}{E\{a^2\}}$



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Solution

(1) SNR, ACF, and PSD of the process $X[n]$.

$$\gamma = E\{S^2[n]\} / E\{W^2[n]\} = E\{A^2\} / \sigma_W^2 = \sigma_A^2 / \sigma_W^2$$

$$R_X[m] = E\{X[n]X[n+m]\} = E\{(A+W[n])(A+W[n+m])\} = E\{A^2\} + R_W[m] = \sigma_A^2 + \sigma_W^2 \delta[m]$$

$$S_X(e^{j2\pi F}) = FT\{R_X[m]\} = FT\{\sigma_A^2\} + FT\{\sigma_W^2 \delta[m]\} = 2\pi\sigma_A^2 \delta(e^{j2\pi F} - 1) + \sigma_W^2$$

(2) Mean vector and covariance matrix of the data vector $\mathbf{X}$.

$$\mathbf{m}_X = E\{\mathbf{X}\} = E\{A\mathbf{1} + \mathbf{W}\} = E\{A\}\mathbf{1} = \sqrt{\frac{\pi}{4}}\sigma_A\mathbf{1}$$

$$\begin{aligned} \mathbf{C}_X &= E\{(\mathbf{X} - \mathbf{m}_X)(\mathbf{X} - \mathbf{m}_X)^T\} = E\left\{\left(\mathbf{X} - \sqrt{\frac{\pi}{4}}\sigma_A\mathbf{1}\right)\left(\mathbf{X} - \sqrt{\frac{\pi}{4}}\sigma_A\mathbf{1}\right)^T\right\} \\ &= E\left\{\left(\left(A - \sqrt{\frac{\pi}{4}}\sigma_A\right)\mathbf{1} + \mathbf{W}\right)\left(\left(A - \sqrt{\frac{\pi}{4}}\sigma_A\right)\mathbf{1} + \mathbf{W}\right)^T\right\} \\ &= \text{var}\{A\}\mathbf{1}\mathbf{1}^T + \mathbf{C}_W = \left(1 - \frac{\pi}{4}\right)\sigma_A^2\mathbf{1}\mathbf{1}^T + \sigma_W^2\mathbf{I} \end{aligned}$$

(3) Maximum A Posteriori (MAP) estimator of A

$$\hat{A}_{MAP} = \arg \max_a f_{A|X}(a|\mathbf{x}) = \arg \max_a f_X(\mathbf{x}, a) = \arg \max_a \ln f_X(\mathbf{x}, a)$$

$$f_X(\mathbf{x}, a) = f_X(\mathbf{x}|a)f_A(a) = f_X(\mathbf{x}|a) \cdot \frac{2a}{\sigma_A^2} e^{-\frac{a^2}{\sigma_A^2}} u(a)$$

where the conditional pdf (in scalar and vector notation) is given by:



$$f_{\mathbf{x}}(\mathbf{x}|a) = \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left(-\frac{(X[n]-a)^2}{2\sigma_w^2}\right) = \frac{1}{\sqrt{(2\pi)^N |\sigma_w^2 \mathbf{I}|}} \exp\left(-\frac{1}{2}(\mathbf{x}-a\mathbf{1})^T (\sigma_w^2 \mathbf{I})^{-1} (\mathbf{x}-a\mathbf{1})\right)$$
$$= \frac{1}{\left(\sqrt{2\pi\sigma_w^2}\right)^N} \exp\left(-\frac{1}{2\sigma_w^2}(\mathbf{x}-a\mathbf{1})^T (\mathbf{x}-a\mathbf{1})\right)$$

Hence, the logarithm of the joint pdf of $\mathbf{X}$ and A can be expressed as:

$$\ln f_{\mathbf{x}}(\mathbf{x}, a) = \frac{N}{2} \ln(2\pi\sigma_w^2) - \frac{1}{2\sigma_w^2}(\mathbf{x}-a\mathbf{1})^T (\mathbf{x}-a\mathbf{1}) + \ln\left(\frac{2}{\sigma_A^2}\right) + \ln(a) - \frac{a^2}{\sigma_A^2}, \quad a > 0$$

that can be rearranged as:

$$\ln f_{\mathbf{x}}(\mathbf{x}, a) = k - \frac{a^2 N}{2\sigma_w^2} + \frac{a \mathbf{1}^T \mathbf{x}}{\sigma_w^2} + \ln(a) - \frac{a^2}{\sigma_A^2} = -\left(\frac{N}{2\sigma_w^2} + \frac{1}{\sigma_A^2}\right) a^2 + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x}\right) a + \ln(a) + k$$
$$= -\frac{N}{2\sigma_w^2} \left(1 + \frac{2}{N\gamma}\right) a^2 + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x}\right) a + \ln(a) + k, \quad a > 0$$

The MAP estimator is obtained by solving the MAP equation:

$$\frac{d \ln f_{\mathbf{x}}(\mathbf{x}, a)}{da} = -\frac{N}{\sigma_w^2} \left(1 + \frac{2}{N\gamma}\right) a + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x}\right) + \frac{1}{a} = 0, \quad a > 0$$

The solution is the positive root of the 2nd order equation:

$$\left(1 + \frac{2}{N\gamma}\right) a^2 - \left(\frac{1}{N} \mathbf{1}^T \mathbf{x}\right) a - \frac{\sigma_w^2}{N} = 0, \quad a > 0$$

Since $\hat{A}_{ML} = \frac{1}{N} \mathbf{1}^T \mathbf{x} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$, the equation to be solved becomes:

$$\left(1 + \frac{2}{N\gamma}\right) a^2 - \hat{A}_{ML} a - \frac{\sigma_w^2}{N} = 0, \quad a > 0$$

and we can express the MAP estimator as a function of the ML estimator as follows:



$$\hat{A}_{MAP} = \frac{\hat{A}_{ML} + \sqrt{\left(\hat{A}_{ML}\right)^2 + 4 \frac{\sigma_w^2}{N} \left(1 + \frac{2}{N\gamma}\right)}}{2 \left(1 + \frac{2}{N\gamma}\right)}$$

When $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (i.e. $\sigma_w^2 \rightarrow 0$) we have: $\hat{A}_{MAP} \rightarrow \frac{\hat{A}_{ML} + \sqrt{\left(\hat{A}_{ML}\right)^2}}{2} = \hat{A}_{ML}$

(4) Bayes Cramér-Rao bound (BCRB)

$$BCRB(A) = - \frac{1}{E \left\{ \frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2} \right\}}$$

$$\frac{d \ln f_X(\mathbf{x}, a)}{da} = - \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} \right) a + \frac{N}{\sigma_w^2} \left(\frac{1}{N} \mathbf{1}^T \mathbf{x} \right) + \frac{1}{a}$$

$$\frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2} = - \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} \right) - \frac{1}{a^2}$$

$$\begin{aligned} -E \left\{ \frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2} \right\} &= E \left\{ \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} \right) + \frac{1}{a^2} \right\} = \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} \right) + E \left\{ \frac{1}{a^2} \right\} \cong \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} \right) + \frac{1}{E \{ a^2 \}} \\ &= \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} \right) + \frac{1}{\sigma_A^2} = \frac{2N}{\sigma_w^2} \left(\frac{1}{2} + \frac{1}{N\gamma} + \frac{\sigma_w^2}{2N} \cdot \frac{1}{\sigma_A^2} \right) = \frac{N}{\sigma_w^2} \left(1 + \frac{2}{N\gamma} + \frac{1}{N\gamma} \right) = \frac{N}{\sigma_w^2} \left(1 + \frac{3}{N\gamma} \right) \end{aligned}$$

$$BCRB(A) = - \frac{1}{-E \left\{ \frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2} \right\}} \cong \frac{\sigma_w^2}{N \left(1 + \frac{3}{N\gamma} \right)} = \frac{N\gamma}{N\gamma + 3} \cdot \frac{\sigma_w^2}{N} = \alpha \cdot MSE \{ \hat{A}_{ML} \},$$

where $\alpha = \frac{N\gamma}{N\gamma + 3}$ and $MSE \{ \hat{A}_{ML} \} = \frac{\sigma_w^2}{N}$. When $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ we have that $\alpha \rightarrow 1$, hence

$$BCRB(A) \rightarrow MSE \{ \hat{A}_{ML} \}.$$

In summary we can say that when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$: $MSE \{ \hat{A}_{MAP} \} \rightarrow MSE \{ \hat{A}_{ML} \} \rightarrow BCRB(A)$.